Cindy Hsu

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Portfolio: https://cindyhsugit.github.io/playground/
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GitHub: https://github.com/cindyhsugit/ai-chatbot-fastapi-with-rag

PROFESSIONAL SUMMARY

GenAI / LLM application engineer with hands-on, production-grade experience building retrieval-augmented generation (RAG) systems. Designed and shipped a two-stage retrieval pipeline (ChromaDB + HuggingFace cross-encoder reranking), a provider-agnostic generation layer supporting OpenAI and Gemini with tiered failover, and a Corrective RAG (CRAG) fallback that grounds answers in live web search when local context and trained knowledge both fall short. Brings 12+ years of production software engineering discipline — version control, testing, deployment, incident debugging — to a deliberate, focused transition into applied GenAI/LLM engineering.

Career note: 2019–2025, extended personal sabbatical (travel); 2025–present, self-directed transition into applied AI/ML engineering.

CORE SKILLS

Python, FastAPI, RAG pipeline design, ChromaDB, FAISS (early-stage prototyping), HuggingFace (sentence-transformers, cross-encoder reranking), OpenAI API, Gemini API, multi-provider LLM integration, retry logic & exponential backoff, async/await, prompt engineering, Docker, Git & GitHub, REST APIs, pytest, cloud deployment (Render, Google Cloud Run), environment variable & secrets management, JSON, command line.

PROJECTS

AI Chatbot with RAG — Python, FastAPI, ChromaDB, HuggingFace, OpenAI API, Gemini API, Docker

Code: github.com/cindyhsugit/ai-chatbot-fastapi-with-rag

- Built and deployed a containerized FastAPI chatbot, live on two independent platforms (Render and Google Cloud Run), each requiring adaptation to its own deployment model.
- Designed a two-stage retrieval architecture: wide-recall vector retrieval (top 20 candidates) followed by HuggingFace cross-encoder reranking (ms-marco-MiniLM-L-6-v2) narrowed to the top 3–5, improving answer relevance beyond raw vector similarity alone.
- Migrated vector storage from a self-managed FAISS index to ChromaDB, gaining built-in persistence and metadata filtering and eliminating custom index-management code; earlier prototyped retrieval directly on raw FAISS to understand vector search fundamentals before adopting a managed vector database.
- Migrated embeddings from OpenAI's paid API to a self-hosted HuggingFace model (all-MiniLM-L6-v2), removing per-call cost and external dependency; diagnosed and resolved a resulting vector dimensionality mismatch (1536 vs. 384) caused by inconsistent embedding usage across synchronous and asynchronous code paths.
- Built a provider-agnostic generation layer supporting OpenAI and Gemini, with a two-tier failover design: provider/network failover (automatic retry on a second LLM provider) and a separate knowledge-coverage failover that detects when neither retrieved context nor the model's trained knowledge can answer a query.
- Implemented a Corrective RAG (CRAG) fallback: when the knowledge-coverage check signals a gap, the system triggers a live web search (Tavily API), grounds a fresh generation call in the retrieved snippets, and flags the answer as web-sourced rather than presenting it as verified local knowledge.
- Verified retrieval correctness through self-directed adversarial testing — injecting fabricated facts into source documents to confirm answers came from retrieved context, not model pretraining — and confirmed correct refusal behavior when a query fell entirely outside the document's scope.
- Resolved real cross-environment issues: a Python version mismatch breaking the Docker build, missing dependencies, and secure credential handling across local, containerized, and production environments.
- Version-controlled with Git across multiple feature branches, including resolving a diverged branch and merge conflict during a history cleanup; wrote unit/integration tests for API endpoints using pytest and TestClient.

PRIOR PROFESSIONAL EXPERIENCE — SOFTWARE ENGINEERING

Senior iOS Developer, StubHub, San Francisco — *March 2017 – May 2019*

- Developed new features for different iOS versions, share extensions, widgets, and public/private libraries.
- Enhanced UI behavior for video/audio call states and responsive settings across iPad and iPhone.
- Improved localization support, increasing non-English speaking users by 300%.

Senior iOS Developer, Udemy, San Francisco — *April 2015 – February 2017*

- Handled build and release processes while collaborating with QA overseas.
- Translated the codebase to Swift and increased Swift usage across the project.

Senior iOS Developer, Mercari, San Francisco — *September 2014 – April 2015*

- Worked with the Japan team to maintain code consistency and support new feature development.
- Tested U.S. and Japan app versions and supported localization testing in English and Japanese.

EDUCATION

M.S. Computer Science — University of Southern California, 2005 – 2007
B.S. Computer Science — Iowa State University, 2001 – 2005